

Concept-Evidence Fusion for Zero-Shot Style Classification

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ABSTRACT Recent advances in vision-language models (VLMs) have established zero-shot image classification as an effective paradigm for recognizing visual categories without task-specific training. Despite their effectiveness in artwork understanding, zero-shot painting style classification remains challenging because visually similar styles can differ only in subtle stylistic characteristics. Existing concept-guided refinement methods use sparse concept memberships to constrain visual neighborhoods, which can leave many paintings without concept-supported neighbors and limit the contribution of concept information to prediction refinement. To address this limitation, we propose Concept-Evidence Fusion (CEF), a training-free framework for transductive zero-shot painting style classification. CEF first converts a dictionary of visual descriptions and named artistic references into direct class evidence through dense image–phrase affinities and class-level text prototypes encoded by a pretrained VLM. Each evidence branch integrates these scores with a visual k -nearest-neighbor graph through label propagation, and the two branches are combined through prediction fusion or cross-modal affinity fusion, with the strategy and parameters selected on labeled validation data. Extensive experiments with four representative VLMs on the WikiArt and MultitaskPainting100k benchmarks with artist-disjoint validation and test splits demonstrate that CEF consistently improves balanced accuracy over vanilla zero-shot inference and existing methods across all eight dataset–backbone combinations without model fine-tuning.

INDEX TERMS zero-shot image classification, painting style classification, vision-language models, concept evidence, score fusion, label propagation.

I. INTRODUCTION

Recent vision-language models (VLMs), such as CLIP [1] and its variants [2]–[4], have achieved strong zero-shot image classification performance by learning aligned representations of images and natural language [5]. By comparing an input image with textual descriptions of candidate categories, these models recognize classes without task-specific supervised training or fine-tuning. Their ability to transfer pretrained representations through text prompts provides a useful foundation for recognition tasks with limited annotated data, including fine-grained artwork understanding.

Painting style classification is an important problem in computational art analysis, supporting digital museums, artwork retrieval and cultural heritage preservation [6], [7]. Recognizing artistic styles is challenging because paintings exhibit subtle stylistic differences, gradual transitions between movements and large variation within each style. Similar subjects may appear across movements, while brushwork, palette and composition distinguish visually related categories. Prior

work has studied supervised painting recognition and art-specific VLM adaptation [8], [9]; frozen VLMs offer an alternative when task-specific training is undesirable. However, class-name prompts alone may not adequately express the distinctions needed for reliable zero-shot style recognition.

Training-free refinement can incorporate relationships among paintings through visual neighborhoods and concept-associated connections [10], [11]. An existing concept-guided approach associates each phrase with a fixed number of paintings and uses their co-membership to constrain a visual graph before uncertainty-gated prediction refinement [11]. This sparse construction can leave many paintings without concept-supported neighbors, requiring a fallback to the unmasked visual graph. Moreover, concept information determines which paintings exchange predictions without directly supplying class evidence. Our controlled analysis shows that the concept mask has little predictive effect under the evaluated settings. These observations motivate a formulation that gives every painting direct access to dictionary evidence.

Paintings within a style often share visual concepts, including characteristic brush strokes, color palettes, compositional layouts and recurring motifs. Nevertheless, many concepts span multiple styles: visible brush strokes occur in Impressionism and Expressionism, while religious subjects appear in Italian Renaissance and Baroque paintings. A shared concept therefore need not identify a reliable path for exchanging class predictions. Dense image–phrase affinities can instead contribute graded evidence to style scores, while class-level text prototypes summarize multiple descriptions. These formulations capture different aspects of the same dictionary and may complement one another. Our dictionary also includes named artistic references; its evidence concerns complete phrases, without isolating descriptive wording from pretrained artist–style associations.

To address these challenges, we propose Concept-Evidence Fusion (CEF), a training-free framework for transductive zero-shot painting style classification. CEF derives phrase-level evidence from dense image–phrase affinities and class-level evidence from CuPL-derived text prototypes, fuses each with VLM scores, and applies label propagation on a graph of images and class nodes. It then combines the branches through prediction fusion or cross-modal affinity fusion, with the strategy and parameters selected on labeled validation data. The framework operates on frozen embeddings and an unlabeled inference pool without model fine-tuning. A follow-up evaluation on WikiArt and MultitaskPainting100k using four representative VLMs and five artist-disjoint splits shows the highest mean balanced accuracy among the evaluated external methods and descriptor-propagation variants on all eight dataset–backbone pairs. Component ablations assess the two evidence formulations and fusion paths; dictionary interventions, regenerated dictionaries and domain probes evaluate the same complete method. Qualitative examples illustrate its prediction changes. These evaluations follow development on previously inspected benchmarks.

Our key contributions are summarized as follows:

- We propose CEF, a training-free framework that combines phrase-level and class-level textual evidence for painting style classification using frozen VLM representations.
- We diagnose the coverage and predictive limitations of sparse concept-guided neighborhoods and formulate direct evidence fusion with validation selection between output and graph combination paths.
- We evaluate CEF across two painting datasets and four VLMs on artist-disjoint splits, with component ablations and additional analyses of dictionary content, regeneration and domain dependence.

II. RELATED WORK

A. VISION-LANGUAGE MODELS AND ARTWORK RECOGNITION

CLIP supports zero-shot classification by comparing images with class-name prompts [1]; MetaCLIP, EVA02-CLIP and SigLIP vary pretraining data, capacity and objectives [2]–[4]. Fine-art style recognition has been studied as a distinct

task [6]–[8], through supervised convolutional models [12], [13], two-stage classification [14], analysis of learned stylistic representations [15], and art-specific vision-language fine-tuning [9]. CEF instead retains the pretrained representation and refines predictions at inference time.

B. CLASSIFICATION BY DESCRIPTION

Menon and Vondrick average similarities to language-generated visual descriptors [16]; CuPL ensembles customized class prompts [17]. LaBo fits an LLM-derived concept bottleneck [18], while other work learns concise descriptor sets [19]. CEF’s phrase-evidence matrix (Equation 8) belongs to this descriptor-classification family. CEF combines this branch with a CuPL-derived prototype branch through output or cross-modal affinity fusion. We therefore compare both standalone descriptor classifiers and controls with propagation, fusion and calibration search (Section V-D).

Random words can recover much of the benefit of LLM descriptors when appended to class names [20]. Removing text alone is therefore insufficient to establish content dependence. Our controls replace standalone phrases with random characters or words and permute their owning styles. Unlike purely visual descriptors, the painting phrases also include artists and works; these controls test complete phrases, not their individual parts.

C. ADAPTATION, CONCEPTS AND SCORE NORMALIZATION

Prompt learning [21], [22], feature adapters [23], support caches [24] and generated support sets [25] adapt pretrained VLMs. Test-time methods adapt per sample [26] or use unlabeled target distributions [27], [28]. CEF shares the transductive use of an unlabeled pool, while adding language-side dictionary evidence and retaining frozen model parameters.

Attribute recognition [29], [30], concept bottlenecks [31] and post-hoc bottlenecks [32] use intermediate concepts for recognition or interpretation; broader VLM developments are reviewed in [5]. CEF’s intermediate quantities are phrase affinities, not supervised attribute measurements. Its optional centering addresses similarity-space hubness [33], following the principle of query-bank and test-distribution normalization [34], [35]; we estimate the offset on validation images and measure its effect separately.

D. GRAPH AND HYPERGRAPH REFINEMENT

Label propagation exploits local and global consistency [36], [37], including large-scale graphs of deep features [38]. Hypergraphs represent higher-order relations for clustering, retrieval and representation learning [39]–[41]. ZLaP connects image and class nodes for training-free zero-shot propagation [10]. CGPR masks visual edges by concept co-membership [11]. CEF adopts ZLaP’s host and changes its cross-modal weights using dense dictionary evidence. The distinction tested here concerns this specific sparse-mask construction and dense alternative, rather than the general usefulness of higher-order graphs.

TABLE 1. Concept coverage at each backbone’s selected configuration. r^* is the selected hyperedge size; “Max cov.” is the ceiling $\min(1, Er/N)$; “Starved” is the fraction of images with no concept-qualified neighbor; “Edges kept” is the fraction of k -NN edges surviving the mask; “Fallback mass” is the fraction of the final graph’s edges contributed by starved rows. These are graph diagnostics on the seed 42 test pool, not repeated accuracy estimates.

Dataset	Backbone	r^*	Max cov.	Starved	Edges kept	Fallback mass
WikiArt	CLIP (OpenAI)	50	12.2%	92.6%	2.2%	97.6%
	MetaCLIP	300	73.3%	64.4%	18.8%	77.4%
	EVA02-CLIP	200	48.9%	74.4%	12.9%	85.2%
	SigLIP	100	24.4%	85.7%	5.6%	93.9%
MultitaskPainting100k	CLIP (OpenAI)	100	14.1%	90.5%	3.3%	96.4%
	MetaCLIP	500	70.4%	64.4%	19.2%	77.1%
	EVA02-CLIP	300	42.2%	75.1%	12.4%	85.8%
	SigLIP	200	28.2%	82.2%	7.7%	91.4%

TABLE 2. Concept-mask diagnosis, balanced accuracy (percent), mean $\pm$ sample standard deviation over five image-level splits. Left: toggle the mask at CGPR’s fixed propagation settings. Right: independently reselect the mask-free uncertainty-gated baseline on each validation split; CGPR retains its original across-seed-selected settings.

Dataset	Backbone	Mask toggled, tuning fixed		Independently re-tuned	
		w/o mask	CGPR	w/o mask	CGPR
WikiArt	CLIP (OpenAI)	47.21 $\pm$ 0.29	47.31 $\pm$ 0.24	47.35 $\pm$ 0.22	47.31 $\pm$ 0.24
	MetaCLIP	45.35 $\pm$ 0.34	45.48 $\pm$ 0.35	45.56 $\pm$ 0.54	45.48 $\pm$ 0.35
	EVA02-CLIP	52.51 $\pm$ 0.18	52.50 $\pm$ 0.28	52.26 $\pm$ 0.38	52.50 $\pm$ 0.28
	SigLIP	48.29 $\pm$ 0.35	48.45 $\pm$ 0.30	48.14 $\pm$ 0.44	48.45 $\pm$ 0.30
MultitaskPainting100k	CLIP (OpenAI)	34.83 $\pm$ 0.12	34.86 $\pm$ 0.12	34.84 $\pm$ 0.12	34.86 $\pm$ 0.12
	MetaCLIP	34.64 $\pm$ 0.15	34.66 $\pm$ 0.11	34.63 $\pm$ 0.14	34.66 $\pm$ 0.11
	EVA02-CLIP	39.33 $\pm$ 0.11	39.33 $\pm$ 0.14	39.38 $\pm$ 0.16	39.33 $\pm$ 0.14
	SigLIP	36.11 $\pm$ 0.22	36.04 $\pm$ 0.23	35.96 $\pm$ 0.18	36.04 $\pm$ 0.23

III. PROBLEM DEFINITION AND MOTIVATING ANALYSIS

A. PROBLEM DEFINITION

Let $\mathcal{D} = \{I_1, I_2, \dots, I_N\}$ denote an unlabeled collection of N paintings, and let $\mathcal{C} = \{c_1, c_2, \dots, c_C\}$ denote the set of C candidate painting styles. Throughout this paper, bold uppercase and bold lowercase symbols denote matrices and vectors, respectively, whereas scalars and individual matrix entries are italic. Given a pretrained VLM with image encoder f_{img} and text encoder f_{text} , painting I_i and style c are represented by the ℓ_2 -normalized embeddings

$$\mathbf{x}_i = f_{\text{img}}(I_i) \in \mathbb{R}^d, \quad \mathbf{w}_c = f_{\text{text}}(p_c) \in \mathbb{R}^d, \quad (1)$$

where p_c denotes the prompt ensemble used to represent style c . Stacking the image embeddings gives $\mathbf{X} \in \mathbb{R}^{N \times d}$.

The vanilla zero-shot probability that painting I_i belongs to style c is

$$Y_{ic} = \frac{\exp(\lambda_{\text{vlm}} \mathbf{x}_i^\top \mathbf{w}_c)}{\sum_{c'=1}^C \exp(\lambda_{\text{vlm}} \mathbf{x}_i^\top \mathbf{w}_{c'})}, \quad (2)$$

where λ_{vlm} is the VLM’s pretrained logit scale. Collecting these probabilities yields the initial prediction matrix $\mathbf{Y} \in [0, 1]^{N \times C}$, and standard zero-shot inference predicts $\arg \max_c Y_{ic}$ independently for each painting.

In addition to the style prompts, an LLM generates a dictionary of E style-associated concept phrases, comprising descriptions and named artistic references (Section V-A3). Let $\mathbf{t}_e \in \mathbb{R}^d$ be the ℓ_2 -normalized VLM text embedding

of concept e , let $s(e) \in \{1, \dots, C\}$ denote its associated style, and let $\mathbf{T} \in \mathbb{R}^{E \times d}$ collect all concept embeddings. The objective is to improve these initial predictions by using both the unlabeled image collection and the concept dictionary without updating the VLM. Labeled validation subsets are used only for model selection, as described in Section V-A. We first test whether concepts are effective as constraints on relationships between images, as assumed by CGPR, and then develop CEF to use them directly as class evidence.

B. CGPR FORMULATION AND ANALYSIS SETUP

CGPR builds a hard incidence matrix $\mathbf{B}$ with entries

$$B_{ie} = \begin{cases} \max(\mathbf{x}_i^\top \mathbf{t}_e, 0) & \text{if } i \in \text{top-}r(e), \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

forms the binary co-membership pattern $\mathbf{M} = \mathbb{I}[\mathbf{B}\mathbf{B}^\top > 0]$, masks a k -NN weight matrix $\mathbf{W}$ with it, and restores the unmasked row for any image left with no surviving neighbor:

$$\widetilde{\mathbf{W}}_{i,:} = \begin{cases} (\mathbf{W} \odot \mathbf{M})_{i,:} & \text{if } (\mathbf{W} \odot \mathbf{M})_{i,:} \neq \mathbf{0}, \\ \mathbf{W}_{i,:} & \text{otherwise (starved node).} \end{cases} \quad (4)$$

Let $\mathbf{D}_W$ be the diagonal matrix of row sums of $\widetilde{\mathbf{W}} + \alpha \mathbf{I}$. Scores are then propagated over $\widetilde{\mathbf{W}} = \mathbf{D}_W^{-1}(\widetilde{\mathbf{W}} + \alpha \mathbf{I})$ with entropy-

based uncertainty gating,

$$u_i = \left(\mathcal{H}(y_i) / \log C \right)^\beta, \quad \mathbf{U} = \text{diag}(\mathbf{u}), \quad (5)$$

$$\mathbf{F}^{(\ell+1)} = (\mathbf{I} - \mathbf{U})\mathbf{F}^{(\ell)} + \mathbf{U}\widehat{\mathbf{W}}\mathbf{F}^{(\ell)}.$$

starting from $\mathbf{F}^{(0)} = \mathbf{Y}$, where y_i is row i of $\mathbf{Y}$ and $\mathbf{u} = [u_1, \dots, u_N]^\top$. Equation 5 is exactly the uncertainty-guided propagation rule of the k -NN method CGPR extends; the concept dictionary enters only through $\widehat{\mathbf{W}}$ in Equation 4.

All measurements in this section use CGPR's own image-level evaluation protocol (Section V-A), because their purpose is to re-examine the results CGPR reported; Section III-D repeats the key comparison on the artist-disjoint splits used for the main results.

C. STRUCTURAL COVERAGE CEILING

Equation 3 selects at most r nonzero memberships per concept, so the number of nonzero entries of $\mathbf{B}$ is at most $E \cdot r$ regardless of N . The fraction of images that can belong to any hyperedge is therefore bounded by $\min(1, Er/N)$. After CGPR's discriminability filter, $E = 40$ for WikiArt ($N = 16,360$ per evaluation partition) and $E = 36$ for MultitaskPainting100k ($N = 25,577$). At $r = 50$, the value selected by CGPR's validation search for CLIP on WikiArt, the ceiling is 12.2%.

Co-membership is a stronger requirement still: an edge (i, j) survives only if i and j are both in the top r of a shared concept. This pairwise condition is more restrictive than individual concept coverage. Table 1 reports what this costs in practice at each backbone's selected configuration. Between 64.4% and 92.6% of images are starved, only 2.2–19.2% of k -NN edges survive the mask, and 77.1–97.6% of the edge mass of the graph actually used for propagation comes from the fallback rows of Equation 4, namely from unmasked k -NN.

Two features of Table 1 are worth separating. The ceiling is a property of the construction: it would hold for a perfect concept dictionary. The starvation rate is worse than the ceiling because concepts retrieve overlapping images and because co-membership is a pairwise condition. The selected r nonetheless spans the grid (50–500), and starvation stays above 64% even at the largest selected value: raising r trades coverage against hyperedge precision, and the validation search never finds a setting where both are adequate.

D. ISOLATING THE CONTRIBUTION OF THE CONCEPT MASK

Table 1 shows the mask is mostly inert structurally; we now measure whether it matters predictively. We hold the propagation hyperparameters (α, K, β) at the values CGPR selected and toggle the mask alone, so the only difference between the two columns of Table 2 is whether Equation 4 is applied. Across all eight (dataset, backbone) pairs the effect is between -0.08 and $+0.15$ balanced-accuracy points, a small effect under the original image-level protocol. On the artist-disjoint splits, toggling the mask at each seed's selected

CGPR configuration changes balanced accuracy by $+0.03$ to $+0.32$ points: the effect is positive on every pair there, but no larger than 0.32 points. Differences are computed from unrounded values; the table reports percentages rounded to two decimals, so recomputing from the printed cells can move an endpoint by one unit in the last digit.

E. CONCEPT-FREE CONTROL

CGPR's original comparison changes both the mask and uncertainty gating. To isolate the mask, we retain Equation 5, use plain k -NN instead of the masked graph, and retune (α, K, β) on the same validation partitions and grid. This concept-free variant removes the dictionary, hyperedges and fallback while preserving the propagation rule.

The right-hand columns of Table 2 report the outcome. Retuned, the concept-free variant lands within -0.30 and $+0.08$ balanced-accuracy points of full CGPR, with no consistent sign, on all eight (dataset, backbone) pairs. Here CGPR uses the configuration of its original search (Section A-D). Against CGPR re-selected on each seed's own validation partition, the concept-free variant is within -0.31 and $+0.10$ points and behind on 7 of eight pairs, so under either selection the concept hypergraph accounts for no more than 0.31 balanced-accuracy points. Together with the left-hand columns, where the mask is toggled with tuning held fixed, this gives the same answer from two directions: under the evaluated settings, the concept hypergraph contributes little to CGPR's improvement. Uncertainty-gated k -NN propagation without the concept dictionary reaches comparable performance.

These measurements revise the attribution of CGPR's gain over non-gated propagation, rather than negating the value of uncertainty gating.

This diagnosis determines the design requirement for the proposed method: concept information must reach every image and must affect class evidence directly, rather than survive the intersection of two sparse neighborhood relations. Section IV implements this requirement.

IV. CONCEPT-EVIDENCE FUSION

A. OVERVIEW

Using the problem definition in Section III-A, we propose Concept-Evidence Fusion (CEF) to incorporate concept information directly into painting-style predictions. CEF uses frozen VLM embeddings, the initial class-name scores and a style-associated concept dictionary. Its two evidence branches retain complementary representations of that dictionary: individual phrase affinities and class-level text prototypes.

The framework proceeds in three stages. First, dense phrase evidence assigns each painting an affinity distribution over the full dictionary and aggregates it by style (Section IV-B). Second, class-prototype evidence compares each painting with normalized averages of class-templated descriptor embeddings (Section IV-C). Third, both types of evidence are combined with class-name scores and integrated through a shared label-propagation graph, either by combining graph affinities before propagation or by combining predictions

Algorithm 1 Concept-Evidence Fusion (CEF)

Require: Labeled validation paintings V , unlabeled test paintings U , style categories $\mathcal{C}$, pretrained VLM, style-owned concept dictionary, search grids (Section V-A)

Ensure: Test prediction scores $\mathbf{F}_{\text{CEF}}$ and predicted styles $\hat{c}_i$

- 1: Extract normalized image and style embeddings; compute class-name predictions $\mathbf{Y}$ (Equation 2)
- 2: Encode standalone phrases and class-templated descriptors
- 3: Average and normalize descriptors within each style to obtain class prototypes
- 4: Estimate optional centering offsets from V only
- 5: **for** each branch $b \in \{\text{p}, \text{u}\}$ (phrase, prototype) **do**
- 6: Construct evidence on V using the branch representation
- 7: Select calibration, class-score fusion and graph settings θ_b on validation bAcc
- 8: Obtain cross-modal affinities $\mathbf{Q}_b$ and propagated validation scores $\mathbf{F}_b$
- 9: **end for**
- 10: Initialize candidate set $\mathcal{K}$ with both original branch predictions
- 11: Add output mixtures for $v \in \{0, 0.1, \dots, 1\}$ using Equation 12
- 12: Add graph mixtures using the two-anchor weight search, graph sweep and weight resweep (Section V-A)
- 13: Select $h^* \in \mathcal{K}$ with highest validation bAcc; break ties in favor of phrase, prototype, then first enumerated mixture
- 14: Freeze h^* , branch settings and centering offsets
- 15: Compute the required evidence on U with frozen settings and offsets
- 16: **if** h^* is an original branch b **then**
- 17: $\mathbf{F}_{\text{CEF}} \leftarrow \mathbf{F}_b$ from its test-pool graph solve
- 18: **else if** h^* is an output mixture **then**
- 19: Solve both test-pool branch graphs
- 20: $\mathbf{F}_{\text{CEF}} \leftarrow (1 - v)\mathcal{N}(\mathbf{F}_p) + v\mathcal{N}(\mathbf{F}_u)$
- 21: **else**
- 22: Form $\mathbf{Q}_{\text{mix}}(v)$ on U (Equation 13)
- 23: $\mathbf{F}_{\text{CEF}} \leftarrow$ scores from the selected fused-graph solve (Equation 11)
- 24: **end if**
- 25: $\hat{c}_i \leftarrow \arg \max_c [\mathbf{F}_{\text{CEF}}]_{ic}$ for each test painting i
- 26: **return** $\mathbf{F}_{\text{CEF}}, \{\hat{c}_i\}$

afterwards (Section IV-D). Validation selects the branch settings and the combination path; inference applies the frozen selection to the unlabeled pool.

Algorithm 1 specifies selection and deployment. CEF’s contribution is the joint use of image-specific phrase evidence and pooled class prototypes within a common propagation framework. The VLM, prototype construction and propagation solver are inherited components; no new parameter-training objective is introduced.

CEF requires no VLM fine-tuning, but its configuration selection uses labeled validation images. Inference is transductive: predictions depend on the other unlabeled images in the evaluation pool. The method therefore assumes an available image collection rather than a strictly isolated test image.

B. DENSE PHRASE EVIDENCE

CEF retains the full dictionary so that each painting can access every style-associated phrase. Optional centering, affinity normalization and class aggregation form a single evidence-construction stage.

We first compute the image–concept similarities $\mathbf{S} = \mathbf{X}\mathbf{T}^T \in \mathbb{R}^{N \times E}$. Some phrases attract high similarities across images. Optional column centering removes this shared offset

before affinity normalization:

$$\tilde{S}_{ie} = S_{ie} - \frac{1}{N_{\text{val}}} \sum_{j \in \text{val}} S_{je}, \quad (6)$$

which removes each concept’s mean similarity over the validation images while preserving how much a given image stands out for it. The statistic uses the validation images but not their labels, and no test image enters it. Whether to apply it is a hyperparameter selected on validation like the others; Section V-F evaluates its effect.

Rather than selecting r member images per concept, we give every image a distribution over the entire concept dictionary. With temperature τ ,

$$A_{ie} = \frac{\exp(\tilde{S}_{ie}/\tau)}{\sum_{e'} \exp(\tilde{S}_{ie'}/\tau)}, \quad (7)$$

When centering is disabled, $\tilde{\mathbf{S}} = \mathbf{S}$. Every row of $\mathbf{A}$ is nonzero: τ controls concentration while preserving full dictionary access. In contrast to top- r selection down each phrase’s similarity column, Equation 7 normalizes each image’s row. High phrase affinity therefore does not require top- r membership.

Each phrase is assigned to the style for which it was generated. This ownership is a dictionary assignment, not a claim that its content is exclusive to that style. The affinity profile therefore induces a second, concept-derived classifier: an image’s evidence for style c is the affinity mass it places on that style’s concepts,

$$Y_{ic}^c = \frac{\sum_{e: s(e)=c} A_{ie}}{\sum_{c'=1}^C \sum_{e: s(e)=c'} A_{ie}}. \quad (8)$$

The resulting $\mathbf{Y}^c \in [0, 1]^{N \times C}$ aggregates complete phrases, including named artistic references.

C. CLASS-PROTOTYPE EVIDENCE

The second branch summarizes the same dictionary at the class level, following CuPL [17]. Each phrase is inserted into the template “{class}, which has {phrase}”, encoded by the frozen text encoder, and averaged with the other templated descriptors of its class. Normalizing this average yields a class prototype; its cosine similarity with each image provides the prototype branch’s class scores. Thus, the phrase branch aggregates image-specific affinities after exponentiation, whereas the prototype branch averages text embeddings before comparing them with images.

Prototype scores enter a calibrated or raw fusion path with the class-name scores. The calibrated path optionally subtracts each class score’s mean over validation images, applies a temperature softmax, and mixes the resulting distribution with the class-name base. The raw path mixes prototype and class-name cosines directly. These are the same dictionary, embeddings and fusion paths used by the CuPL fusion comparator in Section V-D. Both branches select their settings on validation balanced accuracy before their combination is selected.

D. EVIDENCE FUSION AND PREDICTION REFINEMENT

For the phrase branch, we mix the concept-derived distribution $\mathbf{Y}^c$ with the initial class-name prediction:

$$\mathbf{Y}^f = (1 - w) \mathbf{Y} + w \mathbf{Y}^c, \quad (9)$$

with w selected on validation.

Each branch supplies image–class affinities for the same label-propagation host.

Concretely we build ZLaP’s graph [10] over the C class-text nodes and the N images of the unlabeled pool: image i is joined to its k nearest images with weight $\mathbf{x}_i^\top \mathbf{x}_j$, and to its $\min(k, C)$ highest-scoring classes with weight

$$s_{ic} = (Y_{ic}^f)^\gamma \quad \text{in place of} \quad (\mathbf{x}_i^\top \mathbf{w}_c)^\gamma, \quad (10)$$

The fused probabilities are nonnegative. For the raw-cosine fusion base, negative image–class similarities are clamped to zero before the exponentiation.

Validation chooses either the softmax base $\mathbf{Y}$ or raw class cosines; for the latter, $\mathbf{Y}^c$ is scaled by the cosine row maximum. The cosine base at $w = 0$ exactly reproduces ZLaP. Its exhaustively tuned configuration is included among the phrase branch’s candidates, so validation performance cannot be lower by construction; test performance need not follow that ordering. The graph is symmetrized, its diagonal removed, and symmetrically normalized to $\hat{\mathbf{G}}$, and the class scores are the solutions of

$$(\mathbf{I} - \alpha \hat{\mathbf{G}}) \hat{\mathbf{y}}_c = \mathbf{e}_c, \quad c = 1, \dots, C, \quad (11)$$

obtained by conjugate gradients, with image i predicted as $\arg \max_c [\hat{\mathbf{y}}_c]_i$. Apart from the cross-modal weights in Equation 10, this follows the reference label-propagation formulation.

The class nodes supply the propagated labels; fused affinities determine image–class connections rather than being the diffused quantity. ZLaP provides a concept-free reference for the dictionary interventions.

Let $\mathbf{Q}_p$ and $\mathbf{Q}_u$ be the phrase and prototype cross-modal affinity matrices, and $\mathbf{F}_p$ and $\mathbf{F}_u$ their propagated scores. Define $\mathcal{N}$ to clamp entries to zero and divide each row by its sum (floored at 10^{-30}). The output path predicts from

$$\mathbf{H}_{\text{out}}(v) = (1 - v) \mathcal{N}(\mathbf{F}_p) + v \mathcal{N}(\mathbf{F}_u). \quad (12)$$

The graph path instead uses

$$\mathbf{Q}_{\text{mix}}(v) = \text{diag}(\mathbf{m})[(1 - v) \mathcal{N}(\mathbf{Q}_p) + v \mathcal{N}(\mathbf{Q}_u)], \quad (13)$$

where $m_i = \max_c \mathbf{x}_i^\top \mathbf{w}_c$ restores the row scale of the original class cosines. This affinity replaces the cross-modal scores in Equation 10; the graph host remains unchanged.

Validation chooses among the output mixtures, graph mixtures and two original branch predictions using balanced accuracy. These original predictions remain explicit candidates because normalizing graph affinities can change an endpoint’s prediction. The selected configuration is frozen before test inference, and its highest-scoring class gives each painting’s prediction. Section V-A specifies the search grids

and tie-breaking rule; Appendix A-F reports computational requirements and measured cost.

V. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP

All methods share CGPR’s painting pools, frozen VLM representations and reference dictionaries. Unless stated otherwise, evaluation uses five artist-disjoint validation/test splits; original training partitions are unused.

1) Datasets and Evaluation Protocol

WikiArt contains 20,449 paintings in 15 styles from its original validation and test partitions. **MultitaskPainting100k** [8] contains 31,971 paintings in 20 styles after retaining styles with at least 1,000 images, following CGPR. Dataset links appear in Data Availability.

For each seed, we pool the validation and test images and assign each artist wholly to one subset using greedy stratified group assignment in a seed-dependent order, targeting 20% validation per style. Validation contains 20.9–23.2% of images overall and 14.4–32.1% per style, with 136–166 artists on WikiArt and 265–303 on MultitaskPainting100k. Hyperparameters are selected on validation and evaluated on the artist-disjoint test subset. Here, zero-shot refers to the frozen classifier and absence of target-image supervision during inference; selection uses labeled validation images from the test style vocabulary.

CGPR’s image-level protocol instead uses class-stratified image resplits. Across the same five seeds, 95.9–98.6% of test images share artists with validation. We retain this protocol for the CGPR diagnosis and Appendix A-D. The CEF qualitative analysis uses the main artist-disjoint protocol.

2) Vision-Language Model Backbones

We use `open_clip` implementations of CLIP ViT-B/32 [1], MetaCLIP ViT-B/32 [4], EVA02-CLIP ViT-B/16 [3], and SigLIP ViT-B/16 [2], pretrained on WIT-400M, FullCC, Merged-2B and WebLI, respectively. All methods reuse cached embeddings and initial scores $\mathbf{Y}$ from five averaged style-prompt embeddings and the pretrained logit scale (Equation 2).

3) Concept Dictionary Construction

We reuse CGPR’s dictionaries generated by Claude Sonnet 5 (`claude-sonnet-5`), with one request per style and no prompt change except its name.

α : LLM Prompt

The system prompt is “You are an art history expert helping build a visual concept dictionary for zero-shot painting style classification.” The user prompt, reproduced verbatim from the archived generation record, is

Style: “ $\{\text{display_name}\}$ ”

List 10 to 12 short visual concept phrases (3–6 words each) that describe how a viewer could visually recognize

TABLE 3. Main comparison on artist-disjoint validation/test splits. Balanced accuracy in percent, mean $\pm$ sample standard deviation over five seeds for every method. All tunable methods use per-seed validation bAcc selection. DCLIP and CuPL use the same reference dictionary as CEF; their + LP variants use the common label-propagation host with graph tuning. These variants are adaptations for this study. Bold marks the best mean in each row.

Dataset	Backbone	Vanilla	CGPR	ZLaP	DCLIP	CuPL	DCLIP + LP	CuPL + LP	CEF
WikiArt	CLIP (OpenAI)	46.90 $\pm$ 0.25	48.64 $\pm$ 0.26	50.99 $\pm$ 0.60	47.35 $\pm$ 0.30	47.63 $\pm$ 0.19	52.62 $\pm$ 0.47	52.52 $\pm$ 0.53	54.48 $\pm$ 0.32
WikiArt	MetaCLIP	44.55 $\pm$ 0.30	46.09 $\pm$ 0.47	48.25 $\pm$ 0.48	40.46 $\pm$ 0.20	41.52 $\pm$ 0.19	49.58 $\pm$ 0.51	49.98 $\pm$ 0.39	52.03 $\pm$ 0.78
WikiArt	EVA02-CLIP	51.02 $\pm$ 0.13	53.74 $\pm$ 0.28	55.10 $\pm$ 0.57	46.99 $\pm$ 0.18	48.22 $\pm$ 0.12	54.42 $\pm$ 0.53	55.03 $\pm$ 0.50	56.48 $\pm$ 0.69
WikiArt	SigLIP	46.07 $\pm$ 0.25	49.05 $\pm$ 0.27	53.22 $\pm$ 0.59	46.05 $\pm$ 0.34	46.18 $\pm$ 0.36	53.32 $\pm$ 0.39	53.37 $\pm$ 0.26	54.76 $\pm$ 0.98
MP100k	CLIP (OpenAI)	34.27 $\pm$ 0.29	34.78 $\pm$ 0.18	36.24 $\pm$ 0.64	37.76 $\pm$ 0.28	37.04 $\pm$ 0.33	38.97 $\pm$ 0.25	38.80 $\pm$ 0.30	42.09 $\pm$ 0.35
MP100k	MetaCLIP	33.18 $\pm$ 0.52	34.26 $\pm$ 0.47	35.66 $\pm$ 0.65	33.25 $\pm$ 0.41	33.79 $\pm$ 0.39	39.10 $\pm$ 0.49	38.54 $\pm$ 1.08	40.56 $\pm$ 0.46
MP100k	EVA02-CLIP	38.11 $\pm$ 0.42	39.28 $\pm$ 0.53	41.90 $\pm$ 0.75	40.88 $\pm$ 0.54	41.60 $\pm$ 0.52	44.94 $\pm$ 0.45	45.03 $\pm$ 0.66	46.04 $\pm$ 0.40
MP100k	SigLIP	33.66 $\pm$ 0.47	36.35 $\pm$ 0.55	38.85 $\pm$ 0.45	37.85 $\pm$ 0.52	38.06 $\pm$ 0.53	41.21 $\pm$ 0.53	41.36 $\pm$ 0.54	42.27 $\pm$ 0.41

TABLE 4. Evidence contributions and selection controls. Balanced accuracy (percent), mean $\pm$ sample standard deviation over five artist-disjoint splits. Branch selection chooses one of the two CEF branches on validation. DCLIP+CuPL applies the same output/graph combination search to the independently selected DCLIP-fusion and CuPL-fusion branches. All use the reference dictionary.

Dataset	Backbone	w/o prototype	w/o phrase	Branch selection	DCLIP+CuPL	CEF
WikiArt	CLIP (OpenAI)	54.03 $\pm$ 0.67	53.67 $\pm$ 0.35	53.78 $\pm$ 0.45	53.74 $\pm$ 0.29	54.48 $\pm$ 0.32
WikiArt	MetaCLIP	51.69 $\pm$ 0.73	51.04 $\pm$ 0.76	51.69 $\pm$ 0.73	51.29 $\pm$ 0.67	52.03 $\pm$ 0.78
WikiArt	EVA02-CLIP	56.44 $\pm$ 0.79	55.16 $\pm$ 0.67	56.18 $\pm$ 0.92	55.65 $\pm$ 0.90	56.48 $\pm$ 0.69
WikiArt	SigLIP	53.63 $\pm$ 1.25	54.38 $\pm$ 0.75	54.41 $\pm$ 0.76	54.46 $\pm$ 0.83	54.76 $\pm$ 0.98
MP100k	CLIP (OpenAI)	41.67 $\pm$ 0.21	39.62 $\pm$ 0.49	41.67 $\pm$ 0.21	40.15 $\pm$ 0.46	42.09 $\pm$ 0.35
MP100k	MetaCLIP	40.11 $\pm$ 0.37	38.67 $\pm$ 0.68	40.11 $\pm$ 0.37	39.02 $\pm$ 0.48	40.56 $\pm$ 0.46
MP100k	EVA02-CLIP	45.48 $\pm$ 0.63	44.72 $\pm$ 0.75	45.48 $\pm$ 0.63	44.75 $\pm$ 0.54	46.04 $\pm$ 0.40
MP100k	SigLIP	41.41 $\pm$ 0.40	41.57 $\pm$ 0.56	41.41 $\pm$ 0.40	41.54 $\pm$ 0.62	42.27 $\pm$ 0.41

TABLE 5. Comparison with fusion-equipped descriptor baselines on the reference dictionary. Balanced accuracy (percent), mean $\pm$ sample standard deviation over five seeds. Every column uses the same label-propagation host. Descriptor baselines receive score fusion and calibration search, plus exhaustive raw-cosine mixtures. CEF also includes its two-branch combination search. Bold marks the best value in each row.

Dataset	Backbone	ZLaP	DCLIP + fusion	CuPL + fusion	Mean phrase + fusion	CEF
WikiArt	CLIP (OpenAI)	50.99 $\pm$ 0.60	53.59 $\pm$ 0.32	53.67 $\pm$ 0.35	53.51 $\pm$ 0.12	54.48 $\pm$ 0.32
WikiArt	MetaCLIP	48.25 $\pm$ 0.48	51.17 $\pm$ 0.69	51.04 $\pm$ 0.76	50.73 $\pm$ 0.46	52.03 $\pm$ 0.78
WikiArt	EVA02-CLIP	55.10 $\pm$ 0.57	55.45 $\pm$ 0.86	55.16 $\pm$ 0.67	55.20 $\pm$ 0.78	56.48 $\pm$ 0.69
WikiArt	SigLIP	53.22 $\pm$ 0.59	54.35 $\pm$ 0.80	54.38 $\pm$ 0.75	54.05 $\pm$ 0.60	54.76 $\pm$ 0.98
MP100k	CLIP (OpenAI)	36.24 $\pm$ 0.64	40.14 $\pm$ 0.54	39.62 $\pm$ 0.49	40.34 $\pm$ 0.68	42.09 $\pm$ 0.35
MP100k	MetaCLIP	35.66 $\pm$ 0.65	39.07 $\pm$ 0.48	38.67 $\pm$ 0.68	39.33 $\pm$ 0.58	40.56 $\pm$ 0.46
MP100k	EVA02-CLIP	41.90 $\pm$ 0.75	44.97 $\pm$ 0.59	44.72 $\pm$ 0.75	43.58 $\pm$ 0.67	46.04 $\pm$ 0.40
MP100k	SigLIP	38.85 $\pm$ 0.45	41.68 $\pm$ 0.60	41.57 $\pm$ 0.56	41.50 $\pm$ 0.72	42.27 $\pm$ 0.41

a painting in this style, WITHOUT naming the style itself. Cover a mix of:

- brushwork / paint application technique
- color palette / use of light
- composition / form / spatial treatment
- recurring subject matter or motifs
- 1–2 representative artists or landmark works associated with the style (named explicitly)

Do not include the style name or a direct synonym of it in any phrase. Output ONLY a JSON array of strings, nothing else.

The reference dictionaries contain 150 WikiArt and 200 MultitaskPainting100k phrases. CEF, DCLIP and CuPL use all phrases; CGPR's filter retains 40 and 36, respectively, and is tested separately in Section V-F. All phrases, including named references, are embedded intact. Thus the comparison tests dictionary use, without isolating names from descriptive wording. Artist-disjoint image splits do not remove artists from the dictionary or VLM pretraining. Phrases supply in-

ference evidence, never image supervision.

Five further dictionaries per dataset use the same model and prompts, with independent requests, no conversation history and default sampling temperature. They contain 180 and 240 phrases: twelve per style versus ten in the reference dictionary. Every response is archived.

4) Compared Methods

Vanilla VLM predicts from **Y**. **CGPR** [11] uses its reported 400-configuration search space with per-seed validation selection. **ZLaP** [10] propagates class scores on a graph of unlabeled images and class-text nodes; we port and tune the reference implementation.

DCLIP [16] averages image similarities to descriptors templated as "{class}, which has {phrase}". **CuPL** [17] instead compares images with each class's averaged descriptor-embedding prototype. **DCLIP + LP** and **CuPL + LP** replace the common host's cross-modal weights with these scores

and tune its 75 graph settings. CEF combines the phrase and CuPL-fusion branches as specified in Section IV-D; their ablation is in Section V-C. The concept-free CGPR variant is a diagnostic control (Section III-E).

5) Hyperparameter Search

For CEF, both branches and the combination stage use validation bAcc. Historical comparator scores in the image-level protocol retain separate selection for each reported metric. New CEF interventions use bAcc selection. All selections are per backbone and seed. The descriptor, dictionary-control and domain probe analyses use balanced-accuracy selection unless explicitly stated otherwise. For ZLaP, we exhaustively search $k \in \{5, 10, 20, 40, 80\}$, $\gamma \in \{1, 3, 5\}$, and $\alpha \in \{0.3, 0.5, 0.7, 0.9, 0.95\}$, giving 75 graph configurations. This validation search improves balanced accuracy by 2.0 points on average over the published default configuration on the two benchmarks.

The phrase branch searches the same graph parameter space together with the concept temperature $\tau \in \{0.005, 0.01, 0.02, 0.05\}$, the fusion weight $w \in \{0, 0.2, 0.4, 0.6, 0.8\}$, the concept debiasing switch, and the fusion base (softmax probability or raw cosine similarity). The complete Cartesian product contains 6,000 configurations. We therefore use a staged coordinate search that optimizes graph parameters, then evidence parameters, and finally graph parameters again; it evaluates 302 distinct configurations on validation for one metric and 376–452 for both metrics together, including ZLaP’s exhaustive search (Section A-F). The exhaustively selected ZLaP solution is also included among the candidates considered by the phrase branch. As explained in Section IV-D, this ensures that the phrase branch’s selected validation score is not below that of ZLaP.

We use $v \in \{0, 0.1, \dots, 1\}$ for both paths. For output fusion, all weights are scored on validation. For graph fusion, we first evaluate all weights at the two branch-selected graph settings, sweep the 75 graph configurations at the best weight, and resweep weights at the selected graph. The best candidate across both paths and both original branches defines CEF. Ties favor the phrase branch, then CuPL, then the first enumerated mixture candidate. Every dataset, backbone and split follows this same rule; there is no backbone-specific override. All 40 selections were saved and checksummed before CEF test evaluation. Both bAcc and secondary Top-1 results use these same bAcc-selected configurations.

6) Evaluation Metrics

We report Top-1 accuracy and balanced accuracy (bAcc). Balanced accuracy, the mean recall across classes, is treated as the primary metric because both datasets are class imbalanced. CEF and its branch comparisons use validation bAcc for selection and report Top-1 secondarily under the same configurations. The image-level comparator results retain their original per-metric selection where stated; CEF always selects using validation bAcc.

7) Reporting of Uncertainty

Backbones share images within each dataset, and seeds resplit a common pool. Performance tables report means and sample standard deviations ($n - 1$ denominator) across the stated repeated units. These deviations describe observed variability, not confidence intervals. We also report paired effect sizes and consistency rather than treating eight pairs or forty runs as independent replications: mean difference, range across pairs, and positive pair/run counts. Appendix A-G also separates the two datasets. A consistent gain is positive on every pair and nearly every run; sign-changing gains are described as small or unreliable. Tables are generated from archived per-seed outputs, and numerical verifiers check reported comparisons and reproduced reference scores.

B. COMPARISON WITH EXISTING METHODS

Table 3 compares the CEF method with vanilla inference, CGPR, tuned ZLaP, DCLIP and CuPL, including the descriptor classifiers with label propagation. CEF achieves 48.587% mean balanced accuracy, improving on tuned ZLaP by 3.560 percentage points on average. It has the highest mean bAcc among these methods on each of the eight dataset–backbone pairs. All comparisons use the same artist-disjoint partitions and frozen VLM representations.

Balanced accuracy is the primary selection and evaluation criterion. CEF’s secondary Top-1 accuracy under the same selected configurations is 41.369%; Top-1 gains depend on the comparator; Appendix A-A reports the matched bAcc-selected comparison. The secondary comparison with vanilla inference is retained in Appendix A-A.

The remaining experiments examine evidence contributions, dictionary content, validation-label budgets, propagation and domain dependence.

C. COMPONENT CONTRIBUTIONS AND FUSION PATHS

Table 4 examines the contributions of the two evidence branches and their combination. Removing the prototype branch reduces CEF by 0.531 balanced-accuracy points on average; removing the phrase branch reduces it by 1.232 points. Both differences are positive on all eight pair means, ranging from 0.034 to 1.130 and from 0.376 to 2.468 points, respectively; they are positive on 37 and 38 of the forty runs. The small minimum margin over the phrase branch limits the size of the improvement supported by these results.

CEF improves over validation-only branch selection by 0.50 bAcc points on average, with higher means on 8 of eight pairs. This control uses the same selected branches but excludes mixtures. Replacing phrase evidence with DCLIP evidence and applying the same combination search reduces mean bAcc by 1.01 points; CEF is higher on 8 of eight pairs. The alternative retains the independently selected DCLIP/CuPL configurations, including their calibrated and raw searches, and receives the same two-anchor weight/graph/weight combination schedule. Thus, the comparison controls the combination-search procedure rather than assuming that all two-branch ensembles are equivalent.

TABLE 6. Dictionary interventions in CEF. Balanced accuracy (percent), mean $\pm$ sample standard deviation over five seeds. Each intervention changes both the phrase and prototype branches, with both branches and their combination reselected on validation. Counts and phrase lengths are preserved. Prototype templates retain the assigned class name. Bold marks the best mean in each row.

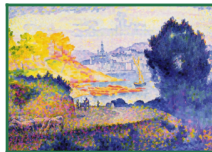
Dataset	Backbone	ZLaP	CEF	Random chars	Random words	Permuted owners
WikiArt	CLIP (OpenAI)	50.99 $\pm$ 0.60	54.48 $\pm$ 0.32	53.14 $\pm$ 0.16	52.69 $\pm$ 0.32	53.17 $\pm$ 0.24
WikiArt	MetaCLIP	48.25 $\pm$ 0.48	52.03 $\pm$ 0.78	49.34 $\pm$ 0.71	49.43 $\pm$ 0.63	49.92 $\pm$ 1.01
WikiArt	EVA02-CLIP	55.10 $\pm$ 0.57	56.48 $\pm$ 0.69	55.19 $\pm$ 0.59	54.95 $\pm$ 0.60	54.96 $\pm$ 0.55
WikiArt	SigLIP	53.22 $\pm$ 0.59	54.76 $\pm$ 0.98	53.31 $\pm$ 0.56	53.27 $\pm$ 0.50	53.67 $\pm$ 0.58
MP100k	CLIP (OpenAI)	36.24 $\pm$ 0.64	42.09 $\pm$ 0.35	37.54 $\pm$ 0.79	37.46 $\pm$ 0.64	37.71 $\pm$ 0.27
MP100k	MetaCLIP	35.66 $\pm$ 0.65	40.56 $\pm$ 0.46	36.05 $\pm$ 0.64	36.03 $\pm$ 0.86	35.76 $\pm$ 0.49
MP100k	EVA02-CLIP	41.90 $\pm$ 0.75	46.04 $\pm$ 0.40	41.72 $\pm$ 0.58	41.87 $\pm$ 0.56	41.81 $\pm$ 0.50
MP100k	SigLIP	38.85 $\pm$ 0.45	42.27 $\pm$ 0.41	38.81 $\pm$ 0.70	38.99 $\pm$ 0.85	38.56 $\pm$ 0.41

TABLE 7. Ablations. Balanced accuracy (percent), mean $\pm$ sample standard deviation over five seeds, with hyperparameters re-selected on validation for each variant. “w/o label propagation” retains both evidence branches and classifies their validation-selected output mixture (Equation 14). Bold marks the best value in each column; ties are marked jointly.

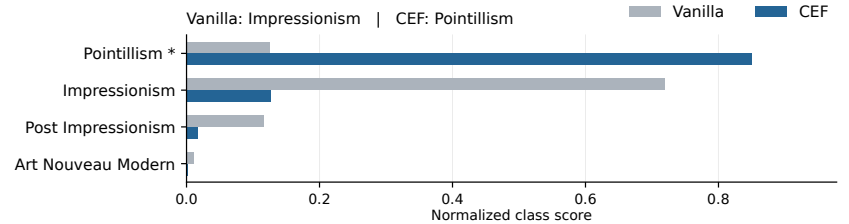
Variant	WA / CLIP	WA / Meta	WA / EVA	WA / Sig	MP / CLIP	MP / Meta	MP / EVA	MP / Sig
CEF	54.48 $\pm$ 0.32	52.03 $\pm$ 0.78	56.48 $\pm$ 0.69	54.76 $\pm$ 0.98	42.09 $\pm$ 0.35	40.56 $\pm$ 0.46	46.04 $\pm$ 0.40	42.27 $\pm$ 0.41
w/o label propagation	51.20 $\pm$ 0.44	48.48 $\pm$ 0.76	53.92 $\pm$ 0.49	51.56 $\pm$ 0.68	39.33 $\pm$ 0.31	39.17 $\pm$ 0.45	44.44 $\pm$ 0.44	40.19 $\pm$ 0.80
w/o centering	54.41 $\pm$ 0.29	52.45 $\pm$ 0.31	56.23 $\pm$ 1.16	54.73 $\pm$ 0.92	41.76 $\pm$ 0.67	40.51 $\pm$ 0.45	46.13 $\pm$ 0.43	42.07 $\pm$ 0.84
w/ filtered dictionary	54.12 $\pm$ 0.36	51.17 $\pm$ 0.37	56.23 $\pm$ 0.44	54.31 $\pm$ 0.53	39.71 $\pm$ 0.16	38.74 $\pm$ 0.52	43.98 $\pm$ 0.79	40.47 $\pm$ 0.80

Artist-disjoint test split | Selected graph fusion: prototype weight 0.3 | * ground truth

(a) Correction



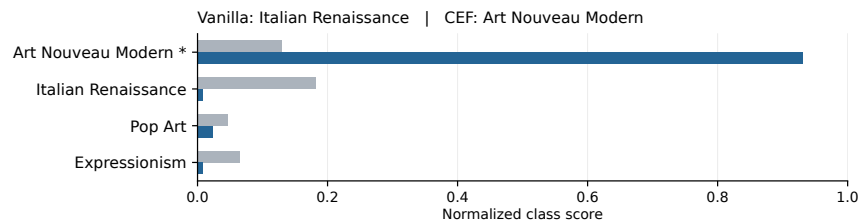
henri edmond cross



(b) Correction



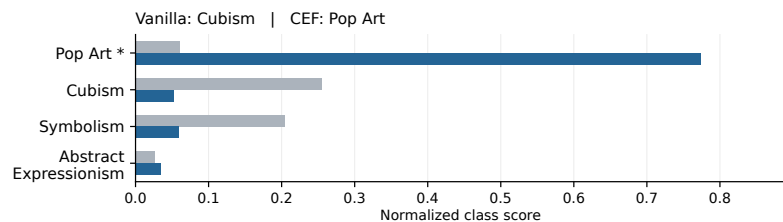
jules cheret



(c) Correction



claudio tozzi



(d) Regression



armand guillaumin

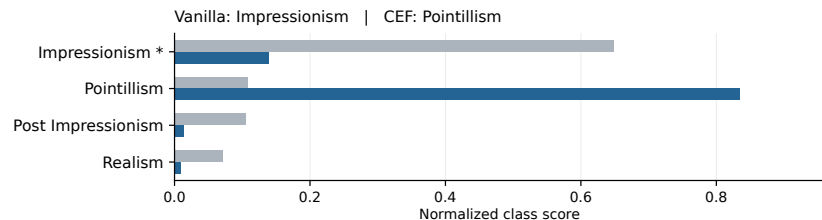


FIGURE 1. CEF on WikiArt, CLIP ViT-B/32, seed 42’s artist-disjoint test split. The frozen validation selector chooses graph fusion with prototype weight $v = 0.3$. Panels (a)–(c) are corrections of vanilla errors; (d) is a regression. Bars show vanilla and CEF scores, normalized over all classes; * marks the true class. The displayed classes include the true label and both predictions. Case selection and the meaning of the scores are specified in Section V-G.

Artist-disjoint test split | Selected graph fusion: prototype weight 0.2 | * ground truth

(a) Correction



(b) Correction



(c) Correction



(d) Regression

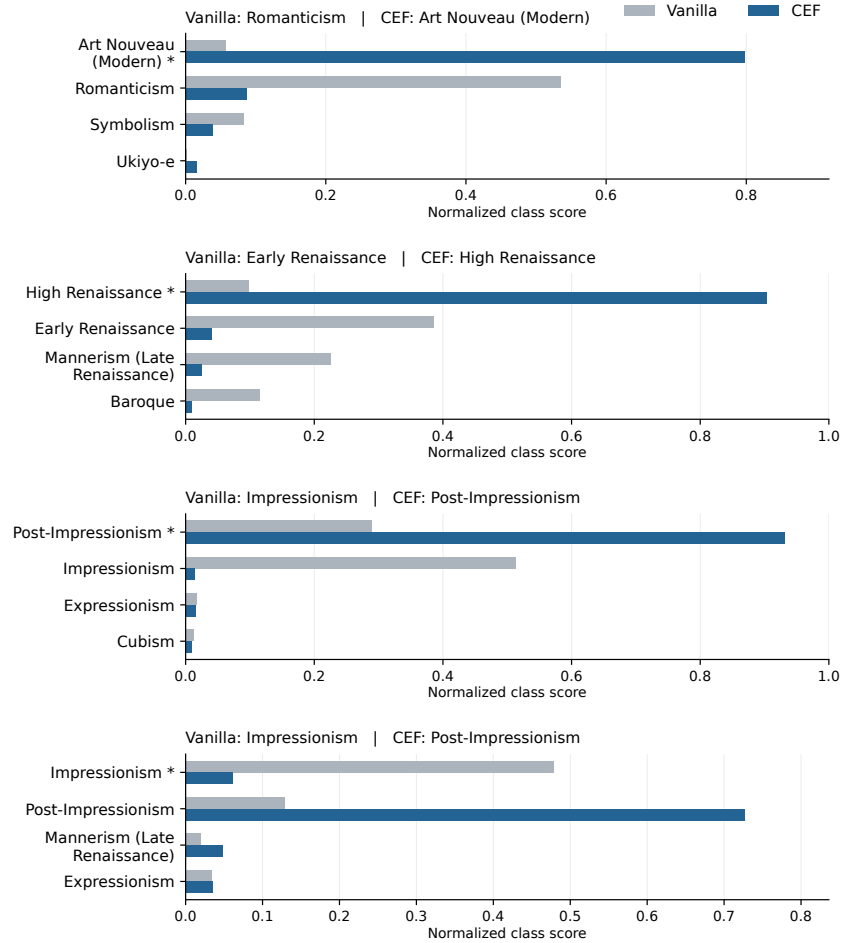


FIGURE 2. CEF on MultitaskPainting100k under the same backbone, seed and artist-disjoint protocol as Figure 1. The frozen selector chooses graph fusion with prototype weight $v = 0.2$. The same case-selection rule and score normalization are used; panels (a)–(c) are corrections and (d) is a regression.

The selector chooses output-mixture candidates on 21 runs, graph-mixture candidates on 18 and an original prototype branch on one. Mixture endpoints can also place zero weight on one branch. CEF therefore makes both evidence sources available but need not use both in every selected prediction. These results establish gains over the evaluated branch and descriptor controls, rather than optimality among all possible fusion rules.

D. COMPARISON WITH FUSION-EQUIPPED DESCRIPTOR BASELINES

Table 5 compares CEF with fusion-equipped descriptor baselines. DCLIP uses mean templated-descriptor cosines; CuPL uses normalized prototype cosines; MeanPhrase averages standalone-phrase cosines within each class. All receive the phrase branch's fusion base, temperature, optional validation centering and weight grids. Centering subtracts each class score's validation mean; softmax at temperature τ yields the fused distribution. MeanPhrase aggregates before exponentiation, while CEF's phrase branch aggregates afterwards.

Each control uses the graph/evidence/graph search and the independently tuned ZLaP candidate, plus all 450 raw-cosine combinations of 75 graph settings and $w \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Validation selects between calibrated and raw paths separately for each metric; $w = 1$ includes descriptor-only LP scores. These baselines include calibration and propagation tuning; CEF additionally searches combinations of the two branches. The runner reproduces all 320 reference metric rows to numerical precision.

CEF exceeds DCLIP, CuPL and MeanPhrase with fusion by 1.0, 1.2 and 1.3 balanced-accuracy points on average, respectively. The differences are positive on all eight dataset-backbone pairs and on 38, 38 and 39 of forty runs. The margins over DCLIP fusion range from 0.4 to 1.9 points.

These results compare CEF with descriptor baselines that also receive score fusion, calibration and graph tuning. CEF additionally selects between combinations of its two evidence branches, so the comparison evaluates the complete methods rather than isolating aggregation under identical search budgets. The branch-removal analysis in Section V-C examines

their contributions within CEF. All results in this comparison use the reference dictionary and the same artist-disjoint splits.

E. CONTRIBUTION OF DICTIONARY CONTENT

Random descriptors can retain gains when appended to class names [20]. Table 6 tests dictionary content within CEF. Random-character and random-word controls preserve each phrase’s word count and its class assignment. The owner-permutation control preserves the original phrases and the number assigned to each class, while permuting those assignments with a fixed random seed. Each intervention affects both branches: the standalone phrase embeddings are used for phrase evidence, and class-templated embeddings are averaged into new prototypes. Both branches and their combination are reselected using the same validation procedure as CEF.

CEF’s mean bAcc differences relative to random characters, random words and permuted owners are 2.9, 3.0 and 2.9 points, respectively. The real dictionary has higher pair means in 8, 8 and 8 of eight comparisons. Unlike a phrase-branch-only intervention, these controls leave no unchanged prototype dictionary that could bypass the perturbed evidence. The class name remains in the prototype template for all controls. The comparison therefore tests the contribution of dictionary content beyond this class-name anchor. It does not distinguish visual descriptions from named artists or works within the original phrases; matched name-only and description-only interventions would be needed for that distinction.

F. EVIDENCE FUSION AND PROPAGATION ABLATIONS

Table 7 evaluates CEF after removing label propagation or optional centering, and after filtering the dictionary. The separate branch-removal comparison is reported once in Section V-C.

For the no-propagation variant, let $\mathbf{Q}_p$ and $\mathbf{Q}_u$ denote the two unpropagated evidence matrices. It predicts

$$\hat{c}_i = \arg \max_c [(1 - \nu)\mathcal{N}(\mathbf{Q}_p) + \nu\mathcal{N}(\mathbf{Q}_u)]_{ic}, \quad (14)$$

with both branches and ν reselected on validation and no graph construction. The mean difference (CEF minus this variant) is 2.5 points, with CEF ahead on 8 of eight pairs. This intervention tests propagation in the complete method rather than removing it from only one constituent.

The filtering intervention applies CGPR’s retained dictionary to both branches and rebuilds the prototypes from the retained descriptors. CEF with the full dictionary differs from this variant by 1.2 points on average, with higher means on 8 of eight pairs. For centering, CEF minus the uncentered variant averages 0.1 points, ranging from -0.4 to 0.3 across pairs. These comparisons concern the evaluated filtering and centering choices, rather than establishing that every possible filter or calibration scheme has the same effect.

G. QUALITATIVE RESULTS

Figures 1 and 2 illustrate how CEF changes vanilla predictions on WikiArt and MultitaskPainting100k, complementing

the quantitative analyses above. Both figures use CLIP ViT-B/32 on the artist-disjoint test splits for seed 42. Validation selects graph fusion with prototype weights of 0.3 for WikiArt and 0.2 for MultitaskPainting100k.

Each figure presents three corrected predictions from distinct ground-truth classes and one initially correct prediction that becomes incorrect. Corrections are selected by the largest increase in the ground-truth class margin over its highest-scoring rival, subject to the distinct-class constraint. The regression is selected by the largest margin decrease, with ties resolved by test index. For visualization and margin computation, negative scores are clipped to zero and scores are normalized over all classes; they are not calibrated probabilities. The examples therefore highlight pronounced prediction changes rather than typical outcomes.

The corrections recover benchmark labels across distinct styles, while the Guillaumin and Renoir regressions illustrate incorrect shifts from Impressionism to neighboring categories. Across all 15810 WikiArt test paintings, CEF corrects 1415 vanilla errors and introduces 816 errors. The corresponding counts for 25015 MultitaskPainting100k paintings are 3046 and 1414, respectively. Thus, corrections outnumber regressions on both evaluated splits, although individual predictions can deteriorate. These cases illustrate the behavior of the combined method without isolating the contributions of individual phrases, prototypes, or visual neighbors.

VI. CONCLUSIONS

In this paper, we proposed Concept-Evidence Fusion (CEF), a training-free framework for transductive zero-shot painting style classification. Unlike existing concept-guided refinement that uses sparse concept memberships primarily to constrain visual neighborhoods, CEF converts concept information into direct class evidence through dense image–phrase affinities and class-level text prototypes. By integrating these complementary evidence sources with visual neighborhood information through label propagation and combining them via validation-selected output or graph fusion, CEF effectively improves zero-shot painting style classification without model fine-tuning. Extensive experiments on the WikiArt and MultitaskPainting100k datasets using four representative vision-language models demonstrated that CEF achieves the highest mean balanced accuracy among the evaluated methods across all eight dataset–backbone combinations, improving over tuned ZLaP by 3.560 percentage points on average. Component analyses further demonstrate the complementary contributions of phrase-level and prototype-level evidence, although improvements in secondary Top-1 accuracy are not uniform across all settings. Although this work focuses on painting style classification, the proposed framework can be extended to other zero-shot visual recognition problems where class-relevant visual concepts can be expressed through language. Future work includes extending CEF to broader recognition tasks and exploring richer concept dictionaries and evidence-fusion strategies.

VII. ACKNOWLEDGMENT

a: Funding

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2026-25495369). This research was supported by Culture, Sports and Tourism R&D Program through the Korea Creative Content Agency grant funded by the Ministry of Culture, Sports and Tourism in 2024 (Project Name: Global Expert Personnel Training in MR Space Computing-based AX Authoring Technology and Immersive Content Development, Project Number: RS-2024-00399186, Contribution Rate: 40%). This work was supported by 2026 Hongik University Innovation Support Program Fund.

b: Use of Artificial Intelligence Tools

Large language models were used to generate visual concept dictionaries, as described in Section V-A3, and to assist with manuscript preparation and analysis-code development. All AI-assisted outputs were reviewed by the authors, who take full responsibility for the content of this work.

APPENDIX A ADDITIONAL CEF EVALUATION

This appendix reports secondary accuracy, dictionary variability, validation requirements, domain probes and computation. All predictions use configurations selected before the corresponding test evaluation.

A. SECONDARY TOP-1 ACCURACY

Table 8 compares Top-1 under the same bAcc-selected configurations. CEF improves on vanilla inference by 4.9 points on average and on all eight pair means. It exceeds ZLaP, DCLIP fusion and CuPL fusion on 8, 7 and 7 of eight pairs, respectively. Thus, the primary bAcc result should not be interpreted as uniform Top-1 superiority over every comparator.

B. VARIABILITY ACROSS DICTIONARY GENERATIONS

Table 9 evaluates five independently generated dictionaries per dataset. Every draw supplies both branches, which are reselected on validation; predictions are not ensembled across draws. CEF exceeds ZLaP by 3.1 bAcc points on average, with higher means on all 40 dictionary-pair combinations. Across-draw standard deviations range from 0.14 to 0.49 points. Reference CEF is 0.5 points higher on average, but its ten phrases per class differ from the twelve in new draws, so this comparison changes phrase count as well as wording.

C. VALIDATION DEPENDENCE AND SOURCE-ONLY SELECTION

Table 10 varies the number of labeled validation images using nested, class-stratified subsets of 10%, 25%, 50% and 100%. The full unlabeled validation pool remains available for graph construction and label-free centering, and test pools are unchanged. Each condition reselects both CEF branches and their combination, and ZLaP's graph, using only subset labels. At 10%, CEF exceeds ZLaP by 1.98 points on WikiArt

and 4.07 on MultitaskPainting100k. The 100% condition exactly reproduces the main CEF results. This experiment isolates annotation budget, not the size of the unlabeled pool.

Table 11 separately removes target-label selection. For each method we take the joint mode of its 80 validation-selected configurations on DTD, EuroSAT, FGVC-Aircraft and Oxford-IIIT Pet (four backbones, five seeds), with lexicographic tie-breaking. CEF's source mode selects its original phrase endpoint with raw class cosines, $w = 0$, and $(k, \gamma, \alpha) = (5, 5, 0.5)$. Both stored branches have these settings; only the selected endpoint executes. Hence this configuration reduces to concept-free ZLaP, and changing the phrase dictionary cannot affect its predictions. Reference and regenerated dictionaries give identical scores. The independently selected source ZLaP uses $(5, 5, 0.3)$.

Source CEF exceeds source ZLaP by 0.94 points on average; this difference is due to graph settings, not dictionary evidence. Target-selected CEF is 4.6 points higher than its source setting. The comparison therefore characterizes these specified source-selection rules and does not show that target validation is unnecessary.

D. COMPARISON UNDER THE IMAGE-LEVEL PROTOCOL

Table 12 uses CGPR's image-level resplits, where validation and test share artists. CEF exceeds ZLaP and CGPR by 3.6 and 6.0 bAcc points on average. CEF selects per seed on bAcc; historical comparators retain their original per-metric selections, and CGPR its original across-seed selection. Artist-disjoint results remain primary. Score differences between protocols change pool composition as well as artist overlap, and cannot isolate leakage.

E. NON-ART DOMAIN PROBES

The probes use Oxford-IIIT Pet [42] (37 classes, 7,390 images), DTD [43] (47, 5,640), EuroSAT [44] (10, 27,000), and FGVC-Aircraft [45] (100, 10,000; copyright banner cropped). Each uses five class prompts, five class-stratified image splits with approximately 20% validation, and the same CEF search. Claude Sonnet 5 generates ten phrases per class using domain-specific visual axes: coat/markings/head shape, texture/pattern, land color/geometry, and aircraft structure, respectively. Prompts prohibit class names and synonyms; Aircraft also prohibits manufacturer, model-family and airline names. Responses containing prohibited class stems or possessives are regenerated. The prototype template subsequently introduces the assigned class name. Prompts and responses are included in the reproducibility archive.

Table 13 compares CEF with vanilla, ZLaP and random words in both branches. CEF-ZLaP differences are 7.3, 6.0, 0.3 and 2.6 points for DTD, EuroSAT, Aircraft and Pet. The corresponding real-minus-random differences are 5.3, 1.4, 0.2 and 0.7. Benefits therefore vary across these domains; the small, dependent probe set does not establish a general predictor of improvement.

TABLE 8. Secondary Top-1 accuracy (percent), mean $\pm$ sample standard deviation over five artist-disjoint splits. All graph methods use their bAcc-selected configurations; no Top-1 reselection is performed.

Dataset	Backbone	Vanilla	ZLaP	DCLIP fusion	CuPL fusion	CEF
WikiArt	CLIP (OpenAI)	40.72 $\pm$ 0.28	39.66 $\pm$ 2.47	42.25 $\pm$ 2.32	42.41 $\pm$ 1.92	44.01 $\pm$ 1.14
WikiArt	MetaCLIP	40.26 $\pm$ 0.29	39.30 $\pm$ 1.33	40.89 $\pm$ 0.49	40.89 $\pm$ 0.32	42.69 $\pm$ 1.01
WikiArt	EVA02-CLIP	45.24 $\pm$ 0.30	45.89 $\pm$ 0.75	45.45 $\pm$ 0.36	45.47 $\pm$ 0.52	46.82 $\pm$ 0.82
WikiArt	SigLIP	36.39 $\pm$ 0.36	41.97 $\pm$ 1.05	43.33 $\pm$ 1.55	43.36 $\pm$ 1.56	43.70 $\pm$ 1.67
MP100k	CLIP (OpenAI)	29.82 $\pm$ 0.25	32.34 $\pm$ 0.55	34.80 $\pm$ 0.63	34.65 $\pm$ 0.81	36.78 $\pm$ 0.53
MP100k	MetaCLIP	31.68 $\pm$ 0.30	32.42 $\pm$ 0.51	34.75 $\pm$ 0.21	34.60 $\pm$ 0.48	36.33 $\pm$ 0.41
MP100k	EVA02-CLIP	36.89 $\pm$ 0.23	39.26 $\pm$ 0.45	40.88 $\pm$ 0.48	41.02 $\pm$ 0.49	42.40 $\pm$ 0.33
MP100k	SigLIP	30.38 $\pm$ 0.31	34.80 $\pm$ 0.60	38.60 $\pm$ 0.54	38.34 $\pm$ 0.68	38.22 $\pm$ 0.49

TABLE 9. CEF across independently generated dictionaries. Balanced accuracy (percent) on artist-disjoint splits. Each draw is evaluated with five seeds. ZLaP and reference CEF report mean $\pm$ sample standard deviation across seeds. The five-draw column gives the mean and sample standard deviation of the five seed-averaged results; the range spans those five means. Each dictionary is used in both branches with validation reselection.

Dataset	Backbone	ZLaP	CEF (reference)	CEF (5 draws)	Draw range
WikiArt	CLIP (OpenAI)	50.99 $\pm$ 0.60	54.48 $\pm$ 0.32	54.13 $\pm$ 0.19	53.80–54.26
WikiArt	MetaCLIP	48.25 $\pm$ 0.48	52.03 $\pm$ 0.78	51.75 $\pm$ 0.25	51.40–52.02
WikiArt	EVA02-CLIP	55.10 $\pm$ 0.57	56.48 $\pm$ 0.69	55.96 $\pm$ 0.14	55.72–56.06
WikiArt	SigLIP	53.22 $\pm$ 0.59	54.76 $\pm$ 0.98	55.25 $\pm$ 0.33	54.96–55.75
MP100k	CLIP (OpenAI)	36.24 $\pm$ 0.64	42.09 $\pm$ 0.35	41.43 $\pm$ 0.19	41.22–41.69
MP100k	MetaCLIP	35.66 $\pm$ 0.65	40.56 $\pm$ 0.46	40.20 $\pm$ 0.44	39.47–40.63
MP100k	EVA02-CLIP	41.90 $\pm$ 0.75	46.04 $\pm$ 0.40	44.72 $\pm$ 0.49	44.02–45.35
MP100k	SigLIP	38.85 $\pm$ 0.45	42.27 $\pm$ 0.41	41.27 $\pm$ 0.24	40.84–41.42

TABLE 10. Validation-label budget. Balanced accuracy (percent), mean $\pm$ sample standard deviation across five seeds after averaging four backbones within each seed. Labeled-image ranges span the five splits. Fractions refer to the existing validation set; unlabeled validation and test pools remain fixed. Both methods reselect their full configurations using only the stated labeled subset.

Dataset	Labels used	Labeled images	ZLaP	CEF
WikiArt	10%	456–481	51.58 $\pm$ 0.45	53.55 $\pm$ 0.26
WikiArt	25%	1127–1191	51.80 $\pm$ 0.33	53.99 $\pm$ 0.27
WikiArt	50%	2246–2376	51.84 $\pm$ 0.31	54.28 $\pm$ 0.25
WikiArt	100%	4488–4747	51.89 $\pm$ 0.38	54.43 $\pm$ 0.31
MP100k	10%	679–724	38.05 $\pm$ 0.38	42.13 $\pm$ 0.62
MP100k	25%	1678–1787	38.16 $\pm$ 0.54	42.25 $\pm$ 0.53
MP100k	50%	3346–3567	38.22 $\pm$ 0.49	42.52 $\pm$ 0.61
MP100k	100%	6679–7125	38.16 $\pm$ 0.56	42.74 $\pm$ 0.36

TABLE 11. Source-only versus target-selected configurations. Balanced accuracy (percent), mean $\pm$ sample standard deviation over five seeds. Each source configuration is the joint mode of 80 non-art validation selections, with no painting-label selection. Target columns use painting validation labels. The selected source CEF is a zero-evidence endpoint, as detailed in the text.

Dataset	Backbone	ZLaP (source)	CEF (source)	ZLaP (target)	CEF (target)
WikiArt	CLIP (OpenAI)	47.51 $\pm$ 0.11	48.98 $\pm$ 0.12	50.99 $\pm$ 0.60	54.48 $\pm$ 0.32
WikiArt	MetaCLIP	45.46 $\pm$ 0.27	46.63 $\pm$ 0.16	48.25 $\pm$ 0.48	52.03 $\pm$ 0.78
WikiArt	EVA02-CLIP	52.77 $\pm$ 0.30	53.75 $\pm$ 0.22	55.10 $\pm$ 0.57	56.48 $\pm$ 0.69
WikiArt	SigLIP	51.47 $\pm$ 0.31	52.41 $\pm$ 0.35	53.22 $\pm$ 0.59	54.76 $\pm$ 0.98
MP100k	CLIP (OpenAI)	34.70 $\pm$ 0.46	35.72 $\pm$ 0.42	36.24 $\pm$ 0.64	42.09 $\pm$ 0.35
MP100k	MetaCLIP	34.53 $\pm$ 0.63	35.16 $\pm$ 0.65	35.66 $\pm$ 0.65	40.56 $\pm$ 0.46
MP100k	EVA02-CLIP	40.92 $\pm$ 0.64	41.59 $\pm$ 0.68	41.90 $\pm$ 0.75	46.04 $\pm$ 0.40
MP100k	SigLIP	37.24 $\pm$ 0.31	37.91 $\pm$ 0.41	38.85 $\pm$ 0.45	42.27 $\pm$ 0.41

F. COMPUTATIONAL COST

Dense phrase affinities cost $O(NEd)$ time and $O(NE)$ storage; class aggregation costs $O(NE)$ and can be batched without constructing an $N \times N$ concept matrix. Embeddings are cached across configurations. After branch selection, graph fusion adds at most 108 validation solves before caching;

output fusion requires 11 score mixtures.

Table 14 measures full post-processing on seed 42, serially in fresh two-thread CPU processes. Selection includes both branch searches and combination selection. Test timing executes only the chosen path: one solve for a graph mixture or endpoint, and two for an output mixture. CEF selection takes

TABLE 12. Image-level resplits with shared validation/test artists. Top-1 and balanced accuracy (percent), mean $\pm$ sample standard deviation over five seeds. CEF selects on validation bAcc and reports Top-1 under that same configuration. Historical comparators retain their original per-metric selections; CGPR uses its original across-seed selection.

Dataset	Backbone	Vanilla		CGPR		ZLaP		CEF	
		Top-1	bAcc	Top-1	bAcc	Top-1	bAcc	Top-1	bAcc
WikiArt	CLIP (OpenAI)	40.38 $\pm$ 0.24	46.25 $\pm$ 0.17	41.93 $\pm$ 0.23	47.31 $\pm$ 0.24	40.35 $\pm$ 0.74	50.57 $\pm$ 0.14	44.22 $\pm$ 0.56	53.58 $\pm$ 0.21
WikiArt	MetaCLIP	40.19 $\pm$ 0.38	44.13 $\pm$ 0.28	41.06 $\pm$ 0.37	45.48 $\pm$ 0.35	40.09 $\pm$ 1.17	48.10 $\pm$ 0.16	42.37 $\pm$ 0.56	51.41 $\pm$ 0.55
WikiArt	EVA02-CLIP	44.81 $\pm$ 0.24	50.30 $\pm$ 0.26	46.64 $\pm$ 0.17	52.50 $\pm$ 0.28	44.64 $\pm$ 0.29	53.95 $\pm$ 0.43	46.42 $\pm$ 0.64	55.60 $\pm$ 0.44
WikiArt	SigLIP	36.26 $\pm$ 0.26	45.87 $\pm$ 0.34	37.51 $\pm$ 0.26	48.45 $\pm$ 0.30	41.89 $\pm$ 0.75	52.78 $\pm$ 0.52	44.02 $\pm$ 0.62	54.85 $\pm$ 0.35
MP100k	CLIP (OpenAI)	29.89 $\pm$ 0.09	34.33 $\pm$ 0.11	30.25 $\pm$ 0.13	34.86 $\pm$ 0.12	31.57 $\pm$ 0.71	36.21 $\pm$ 0.50	36.57 $\pm$ 0.46	41.93 $\pm$ 0.46
MP100k	MetaCLIP	31.87 $\pm$ 0.09	33.34 $\pm$ 0.06	33.43 $\pm$ 0.09	34.66 $\pm$ 0.11	32.57 $\pm$ 0.47	35.78 $\pm$ 0.24	36.47 $\pm$ 0.25	40.74 $\pm$ 0.32
MP100k	EVA02-CLIP	37.21 $\pm$ 0.05	38.48 $\pm$ 0.14	38.32 $\pm$ 0.10	39.33 $\pm$ 0.14	39.30 $\pm$ 0.30	42.05 $\pm$ 0.18	41.66 $\pm$ 0.60	46.11 $\pm$ 0.34
MP100k	SigLIP	30.40 $\pm$ 0.11	33.74 $\pm$ 0.13	32.20 $\pm$ 0.22	36.04 $\pm$ 0.23	34.63 $\pm$ 0.22	38.71 $\pm$ 0.17	39.24 $\pm$ 0.30	42.71 $\pm$ 0.50

TABLE 13. Domain probes of CEF. Balanced accuracy (percent), mean $\pm$ sample standard deviation across five seeds after averaging the four backbones within each seed. Random words replace dictionary phrases in both branches, retaining the prototype class-name template. The last two columns report paired differences in percentage points with their seed-level standard deviations. Painting rows use artist-disjoint splits; non-art rows use image-level resplits.

Dataset	Vanilla	ZLaP	CEF	Random words	CEF – ZLaP	Real – random
WikiArt	47.13 $\pm$ 0.16	51.89 $\pm$ 0.38	54.43 $\pm$ 0.31	52.59 $\pm$ 0.26	2.54 $\pm$ 0.33	1.85 $\pm$ 0.37
MP100k	34.80 $\pm$ 0.38	38.16 $\pm$ 0.56	42.74 $\pm$ 0.36	38.58 $\pm$ 0.66	4.58 $\pm$ 0.24	4.15 $\pm$ 0.49
DTD	53.63 $\pm$ 0.29	55.68 $\pm$ 0.44	62.97 $\pm$ 0.32	57.65 $\pm$ 0.45	7.29 $\pm$ 0.51	5.32 $\pm$ 0.27
EuroSAT	50.94 $\pm$ 0.07	65.66 $\pm$ 0.41	71.70 $\pm$ 0.70	70.28 $\pm$ 0.23	6.04 $\pm$ 0.95	1.42 $\pm$ 0.91
Aircraft	27.06 $\pm$ 0.13	30.42 $\pm$ 0.18	30.69 $\pm$ 0.15	30.54 $\pm$ 0.14	0.26 $\pm$ 0.12	0.15 $\pm$ 0.12
Oxford-IIIT Pet	86.92 $\pm$ 0.15	87.50 $\pm$ 0.12	90.08 $\pm$ 0.19	89.36 $\pm$ 0.19	2.58 $\pm$ 0.17	0.72 $\pm$ 0.18

TABLE 14. CPU post-processing cost for CEF and tuned ZLaP on cached embeddings. Mean $\pm$ sample standard deviation across four backbones on seed 42 artist-disjoint splits, measured serially with two threads per process. CEF selection includes both branches and their combination; test timing executes only the selected path. Peak memory is process RSS, including loaded inputs. The deviations describe backbone variation, not repeated timing noise.

Dataset	Method	Selection (s)	Test (s)	Peak memory (GB)
WikiArt	ZLaP	2.5 $\pm$ 0.1	2.65 $\pm$ 0.29	1.71 $\pm$ 0.02
WikiArt	CEF	33.8 $\pm$ 1.2	2.68 $\pm$ 0.32	1.72 $\pm$ 0.01
MP100k	ZLaP	5.1 $\pm$ 0.1	6.65 $\pm$ 0.52	2.61 $\pm$ 0.03
MP100k	CEF	66.9 $\pm$ 2.7	6.76 $\pm$ 0.58	2.63 $\pm$ 0.02

TABLE 15. Paired bAcc differences (percentage points) for CEF. Means $\pm$ sample standard deviations are computed across five seeds after averaging four backbones. The 95% percentile intervals use 2,000 shared artist-cluster bootstrap resamples per dataset, conditional on the saved predictions and graphs. Positive values favor CEF.

Comparison	WikiArt difference	95% interval	MP100k difference	95% interval
CEF – ZLaP	2.54 $\pm$ 0.33	[2.05, 3.32]	4.58 $\pm$ 0.24	[3.43, 5.42]
CEF – DCLIP fusion	0.80 $\pm$ 0.53	[0.51, 1.23]	1.27 $\pm$ 0.19	[0.60, 1.82]
CEF – CuPL fusion	0.87 $\pm$ 0.45	[0.62, 1.20]	1.59 $\pm$ 0.22	[1.00, 2.09]
CEF – Branch selection	0.42 $\pm$ 0.27	[0.31, 0.56]	0.57 $\pm$ 0.28	[0.40, 0.73]
CEF – DCLIP+CuPL	0.65 $\pm$ 0.48	[0.38, 1.16]	1.38 $\pm$ 0.23	[0.76, 1.90]

33.84 and 66.90 s on the two datasets; inference takes 2.68 and 6.76 s. Peak RSS includes inputs and validation state; loading, VLM encoding and LLM generation are excluded. The reported deviations span backbones, not repeated timing trials.

G. PAIRED UNCERTAINTY

Table 15 reports paired differences from the propagation and descriptor baselines and the two selection controls. Each seed’s difference first averages the four backbones; standard deviations then summarize five seeds. For the 95% intervals,

we resample artists with replacement within each dataset 2,000 times, using the same artist multiplicities across all backbones and overlapping test splits. Each resample recomputes per-class recall and averages bAcc differences over the 20 runs; resamples omitting a class in any split are redrawn. This retains pairing and shared-artist dependence. The intervals hold predictions, selected settings and transductive graphs fixed, and do not include method-selection uncertainty or generalization to new datasets. All displayed intervals exclude zero in favor of CEF under this conditional evaluation.

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